

Just Ask the Model: One-Shot LLM Research Evaluation and Structured Expert Review

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2026-09-07

Abstract

Peer review is strained, and AI tools generating referee-like feedback are already adopted by researchers and commercial services—yet field evidence on how reliably frontier LLMs can evaluate research remains scarce. We compare structured one-shot evaluations by GPT-5 Pro against paid expert review packages from The Unjournal, an open evaluation platform covering economics and social-science working papers, where both humans and the model rate papers on seven percentile criteria with uncertainty intervals and provide narrative critiques. Treating human evaluations as a high-quality but noisy reference signal, we find that GPT-5 Pro approaches the agreement levels observed among human evaluators themselves on several criteria, while exhibiting consistent failure modes: compressed rating scales, uneven criterion coverage, and variable identification of expert-flagged concerns. A minimal one-shot setup—a single prompt with a fixed rubric, no iteration or retrieval augmentation—yields ratings associated with the human panel, supporting further tests of supplementary use. Uncertainty in the human benchmark and assumptions underlying the reliability adjustment prevent an equivalence claim. Appendix results describe five additional models with differing sample coverage.

Introduction

i Choose the right project view

- **Current working paper:** this is the authoritative version of the statistical analysis and academic presentation. It is available at [GitHub Pages](#) and the [primary Netlify URL](#).
- **Full project workspace and grant proposal:** see the [broader project site](#) for project goals, team and funding context, earlier analysis, and appendices. Grant reviewers can go [directly to the proposal](#). This material is retained for context and is not the authoritative source for the latest results.
- **LLM–human critique concordance:** see the [public concordance evidence site](#) for the exploratory issue-mapping judgments and methods notes.

We are also maintaining a small [headless Codex evaluation pilot](#) as an exploratory extension. It is deliberately separated from the main API-based evidence because subscription-backed headless runs are useful for low-cost exploration but are less standardized and less directly replicable than the native API workflow.

*A collaboration with [The Unjournal](#) (55+ open evaluation packages for global-priorities research).
Funding: Survival and Flourishing Fund, Long Term Future Fund, EA Funds.*

Peer review is under strain. Reviewers are hard to find, turnaround times are lengthening, and the system costs an estimated \$1.5 billion per year in the United States alone (Aczel, Szaszi, and Holcombe 2021). At the same time, generative AI lowers the cost of producing polished manuscripts; in at least some fields, editors report submission growth that exceeds reviewer capacity, and explicitly link this trend to LLM-assisted writing (Spitzer 2026). This combination creates demand for automated support in editorial and pre-submission workflows.

Commercial AI reviewer products—such as Refine (Refine n.d.), IsItCredible (IsItCredible.com n.d.), and QED Science (QED Science 2026)—already market automated referee-like feedback directly to authors, though they explicitly disclaim substituting for human peer review and their internal architectures remain undocumented. OpenAIReview (Hsu and Tan 2026) takes a different approach: an open-source, transparent pipeline that uses progressive prompting to generate detailed critiques at roughly \$4 per paper. Meanwhile, publishers are formalizing policies that restrict reviewer use of general-purpose AI tools while permitting controlled in-house applications for screening tasks (Elsevier 2025; Leung 2026). Our study complements these efforts by asking how far the simplest possible setup—a single prompt to a frontier model, with no bespoke pipeline—can go.

These developments make the evidentiary gap salient: funders, editors, and policymakers need to know when AI evaluation outputs are trustworthy enough to use, and when they are unstable, biased, or manipulable. Recent work documents three interlocking concerns. Reproducibility can be “jagged” across models and time (Thomas, Romasanta, and Pujol Priego 2026), and subtle task reframings can induce systematic output shifts reminiscent of specification search (Asher et al. 2026). Adversarial manipulation is not hypothetical: invisible prompt-injection text can inflate LLM review scores in simulated peer review (Choi et al. 2026). And even without manipulation, AI reviews tend to be less thematically diverse and less focused on interpretation and originality than human reviews (Rajakumar et al. 2026), while LLM scoring exhibits range restriction and halo effects that distort agreement metrics (Wang et al. 2025).

The central question we address is therefore: how reliably can frontier LLMs evaluate research, relative to expert peer review and under realistic levels of rater disagreement? We study this question in a setting designed to make “expert judgment” observable and multi-dimensional rather than implicit.

We use The Unjournal’s structured human evaluations as a reference signal. We prompt GPT-5 Pro—a frontier reasoning model—with the same rubric and guidelines used by human evaluators, then compare its ratings with available expert evaluations. The main run covers 60 economics and social-science working papers, with 47 human-matched papers; critique overlap is a separate GPT-5.2 Pro analysis of 12 eligible pairings. We ask whether a frontier LLM evaluation can approximate expert judgment, where systematic differences arise, and whether the model reveals characteristic AI preferences over research. GPT-5 Pro shows positive rank agreement with the human panel, alongside compressed ratings and uneven coverage of expert concerns. A paired bootstrap places uncertainty around an approximate comparison with human inter-rater agreement; it does not establish equivalence to an additional expert. Appendix results describe five additional models spanning different capability and cost tiers, with sample availability varying by model. These findings motivate testing LLMs as supplementary raters in structured evaluation pipelines.

Our approach is minimal by design: each model receives the same PDF and a fixed rubric in a single prompt, with no iteration, retrieval augmentation, chain-of-thought scaffolding, or multi-step agentic loop. This provides a baseline for testing more elaborate evaluation procedures. Structured

measurement schemas (Asirvatham, Moksiki, and Shleifer 2026), iterative quality checks (Zhang and Abernethy 2025), and prompt-sensitivity controls (Asher et al. 2026) could improve performance, but added complexity could also amplify shared errors or introduce new biases. Their incremental value requires comparison on held-out papers.

The Unjournal setting is particularly well suited for this comparison. It commissions paid expert evaluations using a structured rubric covering seven percentile criteria with 90% credible intervals plus journal-tier predictions, and publishes the resulting packages openly rather than making binary accept/reject decisions—which may increase reviewer effort through accountability and transparency. The resulting ratings and critiques still exhibit substantial inter-rater variation; accordingly, we treat human evaluations as a high-quality but noisy reference signal, not ground truth. The rich, multi-dimensional data allow us to compare the priorities and calibration of humans and AI models across criteria and domains, while eventual publication outcomes for journal-tier predictions provide an external validation opportunity¹ enabling a human-vs-LLM horse race. Finally, The Unjournal’s pipeline of future evaluations allows for clean out-of-training-data predictions, serving as a live testing lab for prospective validation.

i Unjournal impact: author engagement evidence

Do authors engage with and respond to Unjournal evaluations? An April 2026 tracking file covers 57 evaluations, but adjustment status had been assessed for 22 rather than the full set. Within that partly assessed snapshot, 16 papers have a recorded formal written response, at least 5 are coded as showing clear evidence of substantive revision, and 7 authors stated an intention to update. A full staff reassessment remains an ongoing goal.

See the [full tabulation: did authors adjust their papers?](#) for the interactive table and public author statements. For broader context, see [Evidence: do authors engage with evaluations?](#) in The Unjournal’s knowledge base.

Results

We evaluate GPT-5 Pro against human expert reviews from [The Unjournal](#) on 47 matched papers (papers with both GPT-5 Pro and human evaluations). The model receives the same PDF, system prompt mirroring The Unjournal rubric, and JSON schema requiring a diagnostic summary plus numeric midpoints and 90% credible intervals for every metric. Results for five additional models are reported in [Appendix A](#). Full methodological details appear in [Methods](#).

We do not treat human ratings as ground truth. Quantitative percentile scoring is genuinely difficult: even domain experts disagree, and individual scores reflect both signal about paper quality and idiosyncratic tendencies (severity, topic familiarity, interpretation of the scale). Our question is whether an LLM provides signal *comparable to an additional expert rater*. The Human–Human baseline row in Table 1 provides this reference. **Caution:** the primary comparison computes the LLM’s ρ against the *mean* of 2.1 human raters, which reduces noise and inflates apparent agreement relative to the individual-vs-individual ρ_{HH} . The Spearman-Brown column provides an approximate sensitivity adjustment. Its classical reliability assumptions (exchangeable raters and independent errors around a common target) need not hold here, and applying it to rank correlations with unequal panel sizes is not an exact correction. Krippendorff’s α_{HH} in the [human-baseline table in Appendix A](#) provides the criterion-level reference.

¹These represent verifiable publication outcomes, not statements about the “true quality” of the paper.

Per-paper overview. Figure 1 presents three complementary views of overall (0–100 percentile) ratings. Panel (a) displays individual human evaluator ratings alongside GPT-5 Pro (orange diamonds) for each paper, revealing inter-rater variability—the self-reported 90% credible intervals from individual evaluators often span 20–40 percentile points. In most cases the LLM falls within the range of human opinions, though several papers show substantial divergence. Panel (b) plots all pairwise human evaluator combinations, showing the observed human-human agreement baseline. Panel (c) compares GPT-5 Pro ratings against human mean ratings with per-paper labels.

Panel (b) displays human-human disagreement, with differences of as much as 30–40 percentile points on individual papers. Panel (c) uses a human mean, so the two scatterplots compare different targets and do not by themselves establish equivalent rater performance. GPT-5 Pro clusters around the identity line in the 40–80 range but diverges more at the extremes, compressing ratings toward the centre of the scale relative to humans—a descriptive pattern whose mechanism is not identified by this comparison. Where humans rate a paper very highly or very harshly, the LLM typically pulls toward the middle. Full agreement metrics for all six models appear in [Appendix A](#).

Rating differences by paper and criterion. Figure 2 unpacks Human – GPT-5 Pro differences across all seven criteria simultaneously. Each column is a paper, each row a criterion, and tile colour encodes the signed difference (human mean minus GPT-5 Pro midpoint). Green tiles indicate the human panel rated the paper higher; orange tiles indicate GPT-5 Pro was more generous. Papers are sorted by overall difference.

Rows with uniformly green or orange tiles indicate papers where humans and GPT-5 Pro disagree systematically across all criteria, not just on one dimension. Columns with consistent colour suggest criteria-level biases—for instance, if “Open Science” is green across nearly all papers, humans may systematically reward data-sharing practices more than the model does. Multi-model comparisons in [Appendix A](#) reveal whether these disagreement patterns are GPT-5 Pro-specific or shared across frontier LLMs.

Qualitative critique comparison. Beyond numeric ratings, we compare the substantive critiques each model raises against the consensus issues identified by human experts. The figure below plots coverage—the fraction of human-identified concerns that the LLM also raised in some form—against the LLM overlap rate—the fraction of LLM-raised issues that correspond to a substantive human concern. These metrics are assessed by GPT-5.2 Pro in a separate judging call (the same model that generated the focal critiques) (see [Methods](#) for the judging protocol). Two of the 14 originally curated pairs are excluded for manuscript-identity mismatches: Peterman et al. 2025 and Bénabou et al. 2023. The latter was identified in the July 2026 concordance audit: its human critiques concern *Willful Ignorance and Moral Behavior*, while the PDF and model critiques concern *Ends versus Means: Kantians, Utilitarians, and Moral Decisions*. The remaining 12 pairs contribute to the analysis. These exclusions concern critique pairings and do not by themselves establish an error in the separate quantitative ratings joins.

Coverage and the LLM overlap rate vary substantially across papers. Averaged over the 12 focal papers, the model captures 80% of human consensus issues (range 50–100%), while 52% of model-raised issues match a substantive human concern. A low LLM overlap rate partly reflects the reference set: the human consensus list contains only the issues evaluation managers prioritised, so model issues flagged by individual evaluators but not curated into the consensus reduce the overlap rate. An unmatched model concern is not necessarily incorrect. Because these metrics are themselves LLM-assessed, they should be interpreted with the caveat that an LLM judge may systematically over- or under-credit matches relative to a human annotator; the repository includes

Figure 1: Per-paper overall ratings (0–100 percentile). **(a)** Individual human evaluator midpoints (green circles) with each evaluator’s self-reported 90% credible interval (reflecting their own uncertainty about the true score; the vertical separation between green dots per paper reflects inter-rater disagreement) and GPT-5 Pro (orange diamonds with CI), sorted by descending human mean. Dotted horizontal lines show grand means. **(b)** Pairwise human evaluator agreement: each point is one evaluator pair (papers with 3 raters contribute 3 points). **(c)** Human mean vs GPT-5 Pro overall rating; dashed diagonal is the identity line. Panel (c) uses an averaged human target, so tighter scatter does not by itself establish equivalence to an individual evaluator.

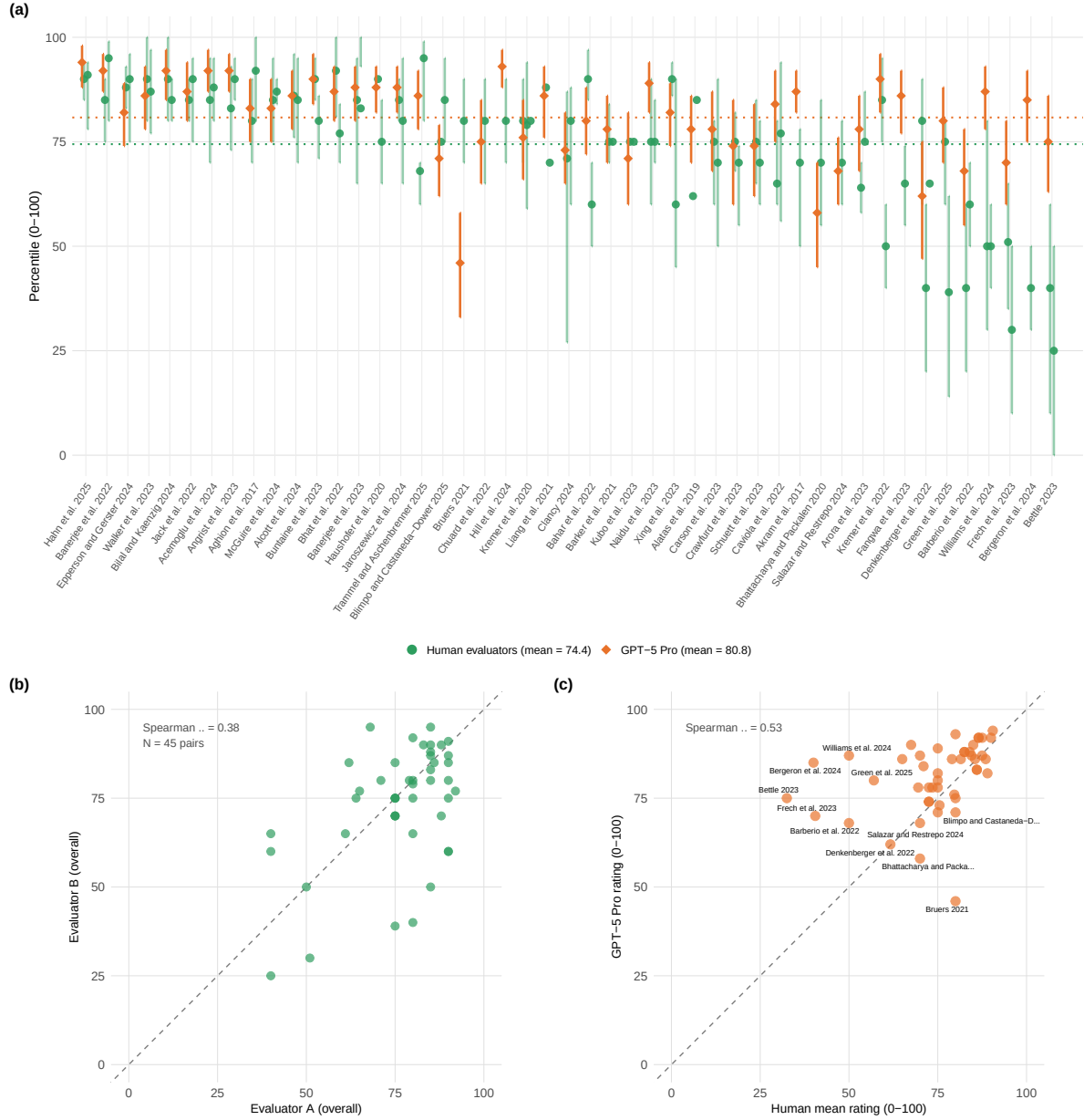
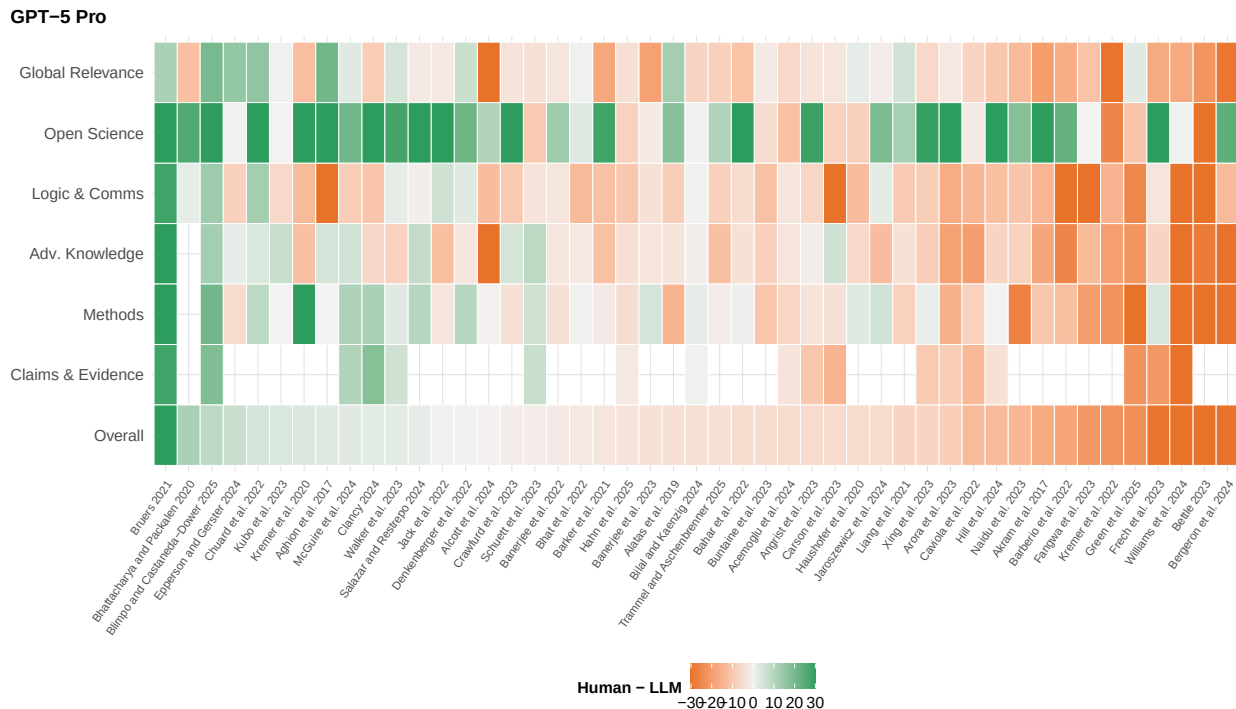


Figure 2: Human minus GPT-5 Pro rating difference for every paper (columns) and criterion (rows). Green tiles indicate the human mean was higher; orange tiles indicate GPT-5 Pro rated the paper higher. Colour is clamped to ± 30 percentile points. Papers are sorted by overall difference.



a browser-based annotation tool for manual validation of these alignment scores, which is ongoing.

Detailed paper-by-paper comparisons—including matched issue pairs with severity labels, structural difference tables, and per-evaluator breakdowns—appear in [Appendix B: Critiques & Key Issues](#). Extended quantitative analysis including all six models, per-criterion correlations, bootstrap confidence intervals, tier prediction accuracy, and cost-quality trade-offs is reported in [Appendix A: Results Ratings](#). The full LLM reasoning traces and assessment summaries are available in [Appendix C: LLM Traces](#).

Agreement summary. Table 1 shows GPT-5 Pro’s agreement split into a **primary matched group** (same 39 papers as the H–H baseline, enabling like-for-like comparison) and a **secondary full-sample group** (all 47 matched papers). In the Spearman ρ column, the H–H row shows individual-vs-individual ρ (the reference, already on the natural scale); GPT-5 Pro rows show ρ vs. the human mean, which is upward-biased. The ρ **adj.** column applies the Spearman-Brown correction to GPT-5 Pro rows only, as an assumption-dependent sensitivity analysis. Six-model comparison in Appendix A, Table A2.

The Fisher-z intervals in Table 1 quantify uncertainty in each correlation separately; to make an uncertainty statement about the *comparison itself* we add a paired bootstrap over papers (2,000 resamples of the 39 matched-group papers). Each draw recomputes the GPT-5 Pro–human-mean rank correlation, applies that draw’s own Spearman-Brown correction, and takes the difference against the human–human pairwise correlation on the same draw. The mean panel size is recomputed within each draw. The estimated 95% interval for the adjusted difference (ρ adj. $- \rho_{HH}$) is $[-0.12, 0.43]$, based on 1965 of 2000 draws. The remaining 35 draws have an undefined correlation

Figure 3: Coverage (% of human-identified issues that the LLM also raised) vs LLM overlap rate (% of LLM-raised issues that match a substantive human concern) for each paper, as assessed by GPT-5.2 Pro acting as judge. Dashed lines mark the cross-paper mean for each axis. Papers in the upper-right quadrant show strong alignment between human and LLM critiques.

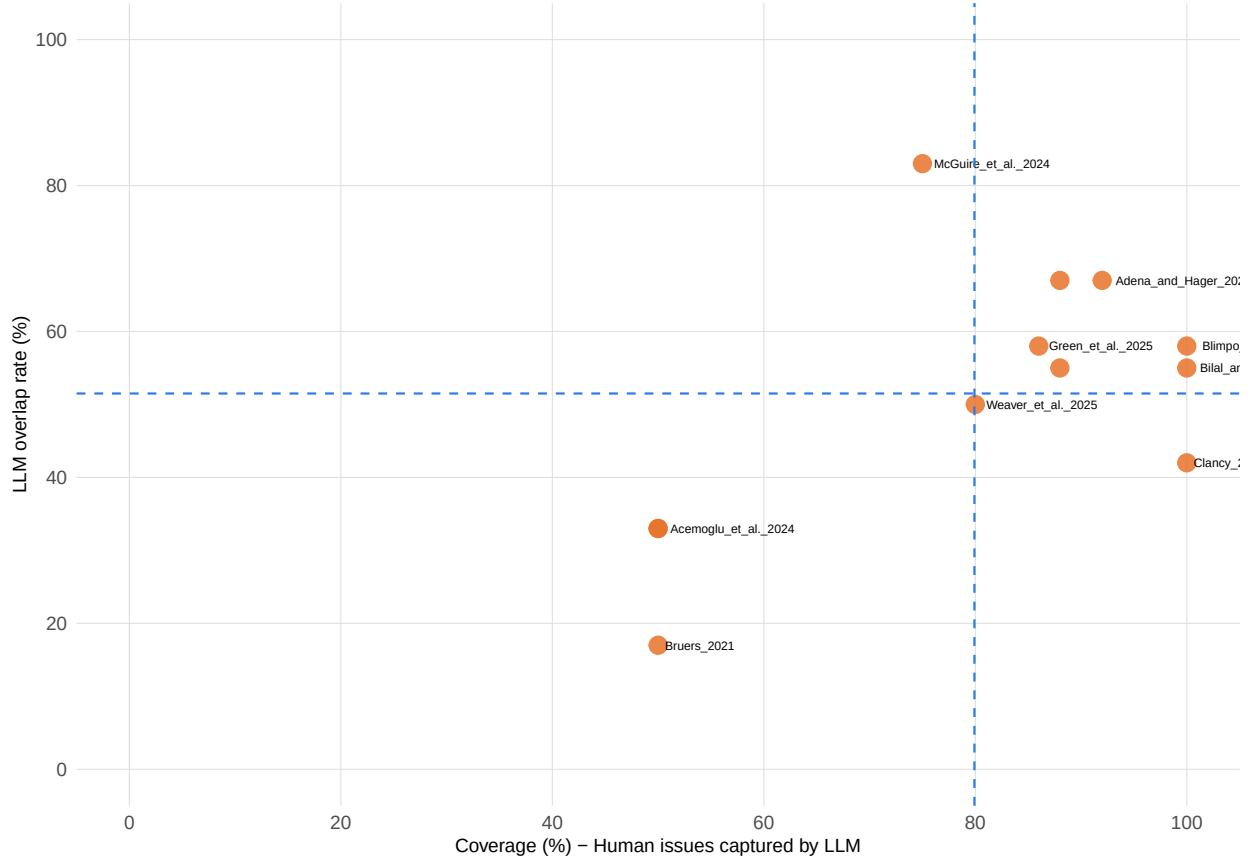


Table 1: GPT-5 Pro overall-rating agreement. **Primary (matched) group:** same 39 papers as the H-H baseline (papers with at least 2 human raters). **Secondary group:** all 47 matched papers. **Spearman rho column:** the H-H row shows individual-vs-individual rank correlation (the reference, already on the natural scale); GPT-5 Pro rows show correlation vs. the human mean, which is upward-biased relative to H-H. **rho adj.:** approximate Spearman-Brown sensitivity adjustment using each row’s mean panel size (matched-group factor 0.81). **95% CI:** approximate Fisher-z interval for the raw correlation, not the adjusted value. Bias = LLM minus Human. Full six-model comparison in [Appendix A](#).

Model	N	Spearman ρ	ρ adj.	95% CI	Pearson r	Mean bias	MAE
Matched sample: same 39 papers as H-H baseline (primary)							
Human-Human	39	0.331	-	[0.02, 0.59]	0.520	+3.7	11.6
GPT-5 Pro	39	0.627	0.508	[0.39, 0.79]	0.577	+6.6	8.4
<i>Full sample: all 47 GPT-5 Pro matched papers (secondary)</i>							
GPT-5 Pro	47	0.526	0.436	[0.28, 0.71]	0.348	+6.4	10.1

Table 2: GPT-5 Pro agreement with human mean by criterion (N = 47 papers; N = 39 for H-H). **H-H rho:** pairwise human evaluator rank correlation (reference; computed on 39 papers with at least 2 raters; raw LLM rho is upward-biased vs. H-H rho, see Table 1). Positive bias = GPT-5 Pro rates higher on average. Note the variation across criteria: human evaluators agree strongly on some dimensions and barely at all on others (Open Science), setting expectations for LLM performance.

Criterion	H-H ρ	Spearman ρ	Pearson r	Mean bias	RMSE
Overall	0.331	0.526	0.348	+6.4	15.1
Claims & Evidence	0.400	0.388	0.213	+4.0	18.3
Methods	0.375	0.562	0.356	+4.5	18.1
Adv. Knowledge	0.148	0.454	0.304	+7.9	17.3
Logic & Comms	0.179	0.287	0.193	+9.5	16.2
Open Science	-0.065	0.150	0.177	-16.8	27.3
Global Relevance	0.114	0.424	0.341	+5.8	15.7

or non-positive human reliability and cannot support this adjustment; the interval is conditional on valid draws. This analysis estimates uncertainty in an approximate comparison. An interval containing zero does not establish equivalence or non-inferiority to an additional expert. Such a claim requires a justified margin specified before testing and an interval narrow enough to rule out a practically important deficit. The present bootstrap also conditions on the observed evaluators and first-two-rater ordering; it does not measure uncertainty from recruiting a new evaluator panel.

Table 2 breaks the same agreement metrics down by evaluation criterion for GPT-5 Pro. The **H-H ρ** column shows pairwise human-human Spearman ρ for each criterion as a reference — note that raw LLM ρ is upward-biased relative to H-H ρ by the mean-vs-individual asymmetry (see Table 1). Criteria where H-H ρ is itself low indicate genuine expert disagreement; these rows alone cannot distinguish reference disagreement from model failure.

Journal-tier predictions and verifiable outcomes. Beyond percentile ratings, both evaluator types predict where each paper *will* publish on The Unjournal’s 0–5 journal-tier scale. Unlike percentile scores, these predictions eventually resolve against an observable outcome: the venue where the paper actually publishes.

Human and GPT-5 Pro tier predictions agree at Spearman $\rho = 0.75$ (N = 42 papers with both predictions)—stronger agreement than on any percentile criterion, suggesting that “where will this publish” is a better-anchored question than “what percentile is this paper.” For 8 of these papers, The Unjournal’s records already show a resolved publication outcome that we can map to the tier scale (Table 3). On this small resolved set, mean absolute error against the realized tier is 0.62 tiers for GPT-5 Pro and 0.71 for the human panel mean—both track outcomes to within roughly two-thirds of a tier, and the sample is far too small to rank the two.

Two caveats apply. First, most of these papers were published before the model’s training cutoff, so GPT-5 Pro may partly *recall* venues rather than predict them; human predictions were made before the publication outcome was known, which makes this comparison structurally generous to the model. Second, the remaining 34 papers with registered predictions from both evaluator types are still unresolved (working papers, R&R, or forthcoming). These predictions are locked in this repository and can be scored as outcomes accrue—a prospective validation design that sidesteps the contamination concern entirely, since future publication decisions cannot appear in any current model’s training data.

Table 3: Papers with resolved publication outcomes: predicted vs realized journal tier (0–5 scale). Realized tier maps The Unjournal’s recorded publication status: ‘~top journal’ = 4.5, ‘decent journal’ = 3.0, field journal = 2.5. Ambiguous statuses, R&R/forthcoming, and book chapters are treated as unresolved and excluded. Human prediction is the evaluator mean.

Paper	Recorded outcome	Realized tier	Human prediction	GPT-5 Pro prediction
Alatas et al. 2019	Published, ~top journal	4.5	3.50	3.50
Barker et al. 2021	Published, ~top journal	4.5	4.00	3.50
Buntaine et al. 2023	Published, ~top journal	4.5	4.70	4.00
Hill et al. 2024	Published, ~top journal	4.5	4.50	4.00
Jack et al. 2022	Published, ~top journal	4.5	4.00	4.00
Fangwa et al. 2023	published, decent journal	3.0	3.60	4.00
Kremer et al. 2020	published, decent journal	3.0	4.00	3.00
Caviola et al. 2022	Published: Psychology journal	2.5	4.35	4.00

All appendices, interactive figures, and full code are available at the companion website: <https://llm-uj-research-eval.netlify.app>. Source code and data are hosted at <https://github.com/valentinklotzbuecher/llm-uj-research-eval>.

Discussion

GPT-5 Pro shows positive rank agreement with human expert evaluations. The comparison with human inter-rater agreement remains uncertain: the paired bootstrap uses an approximate reliability adjustment, and failure to detect a difference does not establish equivalence to an additional expert. Central compression—the tendency to pull extreme ratings toward the middle of the scale—is the most consistent pattern, whose mechanism this observational comparison cannot identify. Journal-tier predictions are the dimension where human and model judgments align most closely, and on the small set of papers whose publication outcomes have already resolved, both predict realized venues with broadly similar accuracy. Qualitative coverage varies widely across papers: on some, the model captures nearly all consensus human concerns; on others, it misses key critiques or raises issues absent from the expert consensus. Appendix results describe five additional models, but differences in paper availability and run configurations limit comparisons across models.

Limitations. Several caveats temper these conclusions. Our sample comprises 60 social-science papers evaluated by the model, of which 47 have published human evaluation packages for direct comparison; all were specifically selected by The Unjournal for evaluation, not a random draw from the research literature; performance may differ in other fields or on less polished manuscripts. Human evaluations are themselves a noisy reference signal rather than ground truth, with substantial inter-rater variation; agreement with a panel mean need not be bounded by pairwise human agreement. We cannot fully rule out knowledge contamination: while we instruct the model to ignore prior knowledge about authors, institutions, or publication history in the system prompt, the models’ training data may include fragments of these papers or related discussions. Robustness checks with models whose training cutoffs predate the papers, or out-of-time validation on papers entering The Unjournal’s pipeline after model training, would help address this concern. Prompt wording, reference-group interpretation, training, and shrinkage toward typical scores are possible explanations for rating patterns; this design does not separate them. Self-reported interval width alone does not establish calibration. All LLM evaluations are single-run; aggregating across multiple runs or temperature settings could change the picture. Finally, the qualitative human-issue coverage and LLM overlap-rate estimates are themselves LLM-assessed (GPT-5.2 Pro as judge),

introducing a further layer of model dependence.

Implications. Even a reasoning-capable model costing several dollars per paper is orders of magnitude cheaper than human expert review. The qualitative gaps we observe—missed critiques, generic issues, and central compression of ratings—argue against full automation of peer review. AI evaluation appears most promising as a supplement: providing fast structured feedback, flagging potential concerns for human reviewers, and enabling systematic comparison across large paper sets that would be infeasible with human effort alone.

Governance and attack surface. As AI review tools move from research prototypes to deployed products, the attack surface expands. Prompt-injection techniques—embedding hidden instructions in a manuscript’s metadata, footnotes, or even white-on-white text—could steer model outputs toward inflated ratings or suppressed critiques. Because our pipeline (and similar commercial services) routes unpublished manuscripts through third-party APIs, confidentiality depends on provider access, retention and training policies, contractual protections, and deployment configuration; encryption in transit alone does not prevent access during inference. Over-reliance on AI scores introduces a further governance risk: if editorial decisions weight model ratings, authors may optimise papers for the model rather than for scientific rigour, creating a Goodhart dynamic. Finally, current evaluations reflect a single model checkpoint; model updates, alignment changes, or fine-tuning can shift ratings in ways that are invisible to users. We recommend that any operational deployment include adversarial red-teaming of prompts, formal confidentiality agreements with API providers, transparent disclosure of AI involvement in review, and periodic re-calibration against fresh human evaluations.

Future directions. All LLM evaluations reported here are single-run; multi-run robustness and prompt-sensitivity analysis would directly address the key open methodological question. Prospective validation should archive the input PDF, model identifier, prompt, and prediction timestamp before human evaluations and publication outcomes become available. Entry into The Unjournal’s pipeline after a training cutoff is insufficient if an earlier manuscript or related review was already public. The qualitative human-issue coverage and LLM overlap-rate estimates are themselves LLM-assessed (GPT-5.2 Pro as judge); human validation of these scores is needed to close the model-dependence loop. Finally, the tier-prediction comparison in Results is a first step toward externally verifiable accuracy measurement: roughly thirty papers with registered predictions from both evaluator types have not yet resolved, and scoring those predictions as publication outcomes accrue can strengthen external validation. Eligibility requires checking that each prediction pre-dates any publicly known outcome or venue cue. A fixed follow-up window and a rule for unresolved papers are needed to avoid selectively scoring early publications.

Methods

This chapter describes the data sources, evaluation pipeline, prompt design, and statistical methods used throughout. All Python code underlying the pipeline is preserved in the collapsible blocks below and can be executed to reproduce the evaluation runs.

Sample and human reference data. We draw on published evaluations from [The Unjournal](#), an open-access platform that commissions expert reviews of policy-relevant research without requiring journal submission. Each paper is typically assessed by two independent evaluators (occasionally one or three), who provide a written critique resembling a referee report together with quantitative ratings on seven criteria scored as percentiles (0–100) relative to “all serious research in the same area encountered in the last three years.”² Evaluators additionally predict the journal tier in which the work “should” and “will” publish, using

²The seven criteria are: *overall assessment, claims and evidence, methods, advancing knowledge and practice, logic*

a 0–5 continuous scale anchored to familiar venue categories (0 = unpublishable, 5 = top-5 journal). For each metric, evaluators report a midpoint (median of their belief distribution) and a 90% credible interval that expresses their epistemic uncertainty.

The sample comprises working papers spanning 2017–2025 in development economics, health policy, environmental economics, and related fields that The Unjournal identified as high-impact. The quantitative human-comparison sample is restricted to papers with published evaluation packages and available matched ratings; the broader model-evaluated set also contains papers without a usable human match. For our analysis we extracted the individual evaluator scores, aggregated them by taking the arithmetic mean per paper per criterion, and retained the range of individual scores to characterise inter-rater spread.

Comparison with an individual evaluator. The main overall-rating table compares the model with the human mean and reports human-human correlation using the first two available evaluator rows on the matched papers. Its Spearman-Brown adjustment is a sensitivity analysis based on classical reliability assumptions, extended approximately to rank correlations and unequal panel sizes. The mean panel size is calculated for the relevant model row and recomputed within each paired bootstrap draw. Draws with non-positive human reliability or undefined correlations are reported as invalid. An interval including zero for the adjusted difference is not an equivalence test. A stronger design would compare the model and a held-out human against the same remaining panel, with a prespecified non-inferiority margin and uncertainty clustered by paper.

LLM evaluation pipeline. We evaluate six frontier models spanning three providers: GPT-5 Pro and GPT-5.2 Pro (OpenAI, reasoning-capable), GPT-4o-mini (OpenAI, lightweight), Claude Sonnet 4 and Claude Opus 4.6 (Anthropic), and Gemini 2.0 Flash (Google). Each model receives the identical PDF file, system prompt, and output schema. We pass the PDF directly to the model’s native multimodal input rather than extracting text, preserving tables, figures, equations, and layout cues that ad-hoc scraping could mangle. A single API call per paper avoids hand-offs and summary loss from multi-stage pipelines.

The output is constrained by a strict JSON Schema enforcing the same nine fields that human evaluators complete: seven percentile metrics (each with `midpoint`, `lower_bound`, `upper_bound`) and two journal-tier predictions (each with `score`, `ci_lower`, `ci_upper`). Additionally, the model produces an `assessment_summary` of approximately 1,000 words that must precede scoring—a “think first, score second” protocol designed to ground numeric ratings in specific textual evidence. The extended schema used for our GPT-5.2 Pro focal run adds a `key_issues` array of concise, ranked issue statements for downstream critique comparison.

Agreement and reliability statistics. We quantify human–LLM agreement using several complementary measures. *Pearson’s r* captures linear association between paired ratings; it is invariant to positive linear rescaling and constant offsets, so it does not detect all forms of rating bias. *Spearman’s ρ* measures rank-order agreement and is robust to outliers and non-linear monotone relationships. *Mean bias* (LLM minus human) indicates the direction and magnitude of any systematic offset. *Root mean squared error* (RMSE) and *mean absolute error* (MAE) measure typical prediction error in the original scale units; RMSE penalises large deviations more heavily. To contextualise human–LLM agreement we report *Krippendorff’s alpha* (α), a chance-corrected reliability coefficient that generalises across varying numbers of raters, accommodates missing data, and applies to any measurement level (nominal, ordinal, interval, or ratio). An α of 1 indicates perfect agreement, 0 indicates agreement no better than chance, and values below 0 indicate systematic disagreement. We compute α_{HH} (among human evaluators only) as a descriptive reference. The appendix also reports α_{HL} for the panel formed by adding each LLM to the individual human evaluators. This describes the reliability of the augmented panel, not a stand-alone model-human coefficient. The reference is not a mathematical ceiling, and a similar augmented-panel value does not establish that a model is interchangeable with a human evaluator. Model availability varies within the appendix paper set. For the qualitative key-issue comparison, *coverage* denotes the fraction of human-identified issues that received a match score $\geq 30\%$ from the LLM judge. The *LLM overlap rate* denotes the fraction of LLM-generated issues that matched at least one human issue; unmatched LLM concerns are not necessarily incorrect.

and communication, open and collaborative science, and relevance to global priorities. Full definitions are given in The Unjournal’s [guidelines for evaluators](#).

JSON Schema. The output schema enforces that every paper is scored on identical fields with identical types and bounds. Credible intervals are required (paralleling the human protocol) so that the model can express genuine uncertainty rather than suggest false precision.

System prompt design. The system prompt is assembled from modular components and concatenated before each API call. It opens with a role definition instructing the model to act as an expert research evaluator, followed by a debiasing block that explicitly prohibits use of author identity, institutional prestige, publication venue, or any extrinsic information—the model must base all judgments on the PDF content alone. A diagnostic-summary instruction requires the model to produce a roughly 1,000-word assessment identifying methodological, evidential, and interpretive issues *before* any scoring, implementing a “think first, score second” protocol intended to anchor numeric ratings in specific textual evidence.

Percentile ratings are anchored to the reference group “all serious research in the same area encountered in the last three years,” following The Unjournal’s [guidelines for evaluators](#). The prompt defines each of the seven criteria with emphasis on global priorities and practical relevance over pure academic novelty, mirroring the weight structure that human Unjournal evaluators are asked to apply.

Subsequent prompt components instruct the model on constructing 90% credible intervals—the smallest interval the evaluator believes is 90% likely to contain the true value—encouraging calibrated uncertainty rather than artificially narrow bounds. The prompt requests journal-tier predictions on a 0–5 continuous scale anchored to familiar venue categories (0 = unpublishable through 5 = top-5 journal), providing an externally verifiable reference point for papers that eventually publish. A validation block then requires the model to verify internal consistency: numeric scores must align with the written assessment, credible intervals must be non-degenerate, and high or low ratings must be explicitly justified in the assessment summary.

The components above are concatenated into a single system prompt string before each API call:

Submission and collection. The `evaluate_paper` function uploads a PDF to the API and submits a background job for evaluation. File IDs are cached by path, size, and modification time so that re-running on the same PDF reuses the previously uploaded file.

Each model receives the full PDF in a single reasoning call, avoiding hand-offs and summary loss from multi-stage ingestion. We submit one background job per paper to the OpenAI Responses API with “high” reasoning effort and server-side JSON-Schema enforcement, recording the response ID, model, file ID, status, and timestamps in a jobs index. No external sources or cross-paper material are retrieved; the evaluation is anchored entirely in the manuscript itself.

A polling loop then checks each job’s status and, for completed jobs, retrieves the raw JSON response object and writes it to disk alongside reasoning-trace metadata (token counts, reasoning summary).

Multi-model evaluation. To assess whether systematic biases or calibration differences emerge across architectures, we collect evaluations from six models spanning three providers. For OpenAI models that lack background-job or extended-reasoning support (e.g., GPT-4o-mini), we use a synchronous call variant:

Anthropic. Anthropic’s API accepts PDFs as base64-encoded document content rather than uploaded file IDs, and all calls are synchronous. We use Claude’s native PDF support to preserve the same multimodal evaluation approach:

Google. Google’s Gemini API accepts PDFs via a file-upload endpoint with MIME-type tagging:

A unified runner dispatches evaluations across all configured providers and writes per-model output directories:

Focal run with key-issue extraction. For a subset of 14 papers with rich human critiques, we ran an extended evaluation using GPT-5.2 Pro. The schema for this focal run adds a `key_issues` array—a ranked list of concise issue statements identifying the most important methodological, interpretive, or evidential concerns—alongside the standard metrics. This structured output enables direct comparison between machine-generated and human-identified issues. Two curated pairings (Peterman et al. 2025 and

Bénabou et al. 2023) were subsequently excluded because their human and model critiques concerned different manuscripts; the coverage/overlap analysis therefore covers 12 papers. The Bénabou mismatch was identified in the July 2026 concordance audit. Raw responses are retained, with exclusions applied consistently in both results chapters.

Key-issue comparison with human critiques. To validate how well the LLM identifies substantive concerns, we compare its `key_issues` output against human expert critiques drawn from The Unjournal’s Coda database. These human critiques—produced by paid domain experts and synthesized by evaluation managers—provide a high-quality but noisy reference standard for issue identification; accordingly, we treat them as a comparative signal rather than ground truth.

The comparison proceeds in two stages. First, we parse and align the data sources: LLM key issues are extracted from the focal-run JSON responses, while human critiques are drawn from a manually curated markdown document pairing each paper’s machine and expert assessments. Second, we use an LLM judge (GPT-5.2 Pro with schema-enforced structured output) to systematically assess the degree of alignment between each human-identified issue and the set of machine-generated issues, producing issue-by-issue match scores, coverage estimates (fraction of human issues captured), and LLM overlap-rate estimates (fraction of LLM issues that match the curated human concern set).

The comparison pipeline takes a manually curated markdown file pairing each paper’s LLM issues with human critiques and produces two structured outputs: a parsed JSON dataset (`key_issues_comparison.json`) and an LLM-assessed alignment report (`key_issues_comparison_results.json`) containing per-paper coverage, LLM overlap rate, matched-pair explanations, and an overall quality rating. Together, these outputs feed the qualitative analysis presented in [Appendix B: Critiques & Key Issues](#).

Headless Codex Pilot

This page records a small exploratory extension to the main evaluation pipeline: running the same broad research-evaluation scaffold through `codex exec` using ChatGPT/Codex subscription access rather than metered OpenAI API calls.

This is not part of the main evidence base. The primary results in this working paper use API-based model calls with controlled inputs, recorded model identifiers, and native PDF ingestion where available. The headless Codex pilot is useful for low-cost exploration, prompt development, and deciding whether further evaluations are worth running, but it should be treated as a separate input modality and execution surface.

What we tried

We added a small runner:

- Script: `scripts/run_headless_codex_eval.py`
- Pilot run: `results/codex_gpt55_headless_pilot_jul2026/`
- Model surface: `gpt-5.5` via `codex exec`, authenticated through ChatGPT/Codex rather than an API key
- Prompt version: `v4_assessment_current_text_extraction`
- Input modality: `pdftotext`-extracted text, truncated at 60,000 characters for the first pilot
- First paper: `Bruers_2021.pdf`

The runner deliberately marks the result as `pdftotext_extracted_text`, stores the source PDF hash, writes a `run_config.json`, and emits JSON in a compatibility shape that the existing analysis code can read. The aggregation helper now reads both older OpenAI response payloads and the newer `parsed / output_text` shape used by this pilot.

The one-paper pilot produced the expected artifacts:

- `json/Bruers_2021.response.json`
- `metrics_long.csv`
- `tiers_long.csv`
- `combined_long.csv`
- `assessment_summaries.csv`

For this first run, the combined metrics were:

Metric	Midpoint	Lower	Upper
Overall	46	32	61
Claims and evidence	42	27	58
Methods	34	20	50
Advancing knowledge	57	40	72
Logic and communication	62	48	75
Open science	36	20	55
Global relevance	70	55	82
Tier should	2.3	1.5	3.2
Tier will	2.4	1.4	3.4

The model also correctly noted in its assessment that the text had been truncated and widened uncertainty accordingly. This is the behavior we want from an exploratory runner: it should surface input limitations rather than presenting the output as equivalent to a full native-PDF evaluation.

Terra → Sol concordance follow-up

We also tested a two-stage concordance workflow for comparing separately enumerated human issues with LLM-generated issues. A high-effort Terra pass checked input alignment and proposed mappings; only three ambiguous pair judgments were escalated to a high-effort Sol pass. The workflow used stable issue IDs, retained a historical overlap threshold of 30, recorded exact Codex telemetry, and kept all outputs separate from the confirmatory API results.

We fixed the candidate set as the first three records in source order before inspecting concordance. A manual title/PDF/issue screen excluded Bénabou et al. before judgment: the recorded title and human issues concerned *Willful Ignorance and Moral Behavior*, while the local PDF and LLM issues concerned *Ends versus Means: Kantians, Utilitarians, and Moral Decisions*. Acemoglu et al. and Adena–Hager passed the screen and were processed together in one Terra batch.

Result	Acemoglu et al.	Adena–Hager	Pooled eligible issues
Human-issue coverage	2/3 (66.7%)	6/7 (85.7%)	8/10 (80.0%)
LLM overlap rate	3/12 (25.0%)	6/12 (50.0%)	9/24 (37.5%)

The LLM overlap rate is the fraction of unique LLM issue IDs with at least one non-distinct mapping scoring 30 or higher. It is not an estimate of whether unmatched LLM concerns are incorrect. Sol changed one threshold-sensitive mapping out of the three escalated pairs: an Adena–Hager mapping moved from 31 to 27, lowering the pooled LLM overlap rate from 41.7% to 37.5%. Human-issue coverage was unchanged because that human issue retained another qualifying match.

The original Terra and Sol calls used 41,993 input tokens and 4,001 output tokens, including 1,229 reasoning-output tokens. An identical verification run reused both completed artifacts and made

no new model calls. The compact [methods note](#), [eligibility screen](#), and [structured final judgments](#) are retained in the repository.

This increment supports preparing a larger preregistered-style study, but not running the full corpus yet. The remaining inputs should first receive the same alignment audit, and the protocol should freeze eligibility, compound-issue handling, aggregation, ambiguity ranking under an escalation cap, and treatment of ambiguous pairs that are not escalated.

Replicability and consistency

The main concern is real: this approach is less standardized than the API-based approach.

The API workflow is better for strict replication because it gives us a cleaner contract: explicit model identifiers, structured-output enforcement, direct request and response payloads, file IDs or native PDF inputs, usage metadata, and a provider surface designed for programmatic runs. It is also easier to re-run from a clean environment, pin defaults, and audit failures.

The headless Codex workflow is attractive for exploration because it can use subscription access and avoid metered API costs. But several details make it less directly replicable:

- The execution surface is the Codex CLI, not the model API.
- Subscription entitlements and rate limits can vary by account and over time.
- Default behavior may change with Codex releases unless we pin command flags and record the CLI version.
- The pilot currently uses extracted text, not native PDF input, so figures, tables, equations, and layout can be degraded.
- The first run intentionally truncated the extracted text; this must be treated as a pilot constraint, not a production setting.
- `codex exec` is an agentic wrapper around a model call. Even when prompted not to use tools, it is not as minimal as a direct API request.

This does not make the approach unusable. It means we should use it with a different evidential status. It is well suited for:

- cheap preliminary screening;
- prompt iteration before API spending;
- exploratory robustness checks;
- identifying candidate papers for deeper API-based evaluation;
- running larger informal batches where exact comparability is not the goal.

It is less suited for:

- headline quantitative comparisons with human evaluators;
- claims about model-vs-model performance;
- strict reproduction of the native-PDF GPT-5 Pro runs;
- archival baselines where future re-runs must be as close as possible to the original call.

Practical guardrails

For any larger headless batch, we should keep the following guardrails:

- Record the Codex CLI version, model flag, auth mode, run timestamp, prompt version, PDF hash, text-extraction command, input character count, and truncation status.
- Keep the output in a separate `results/<run_id>/` directory with `input_modality` clearly marked.

- Do not merge headless/text-extraction results into the main model-comparison plots without a visible modality label.
- Use API-based runs for confirmatory comparisons or for any result we expect others to reproduce exactly.
- Avoid browser automation or ChatGPT web scraping; use documented Codex non-interactive execution instead.

The appropriate interpretation is therefore: headless Codex is a useful exploration and cost-control tool, not a drop-in replacement for the API-based evidence pipeline.

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Results: Ratings

Extended ratings analysis (per-criterion correlations, agreement metrics, human baseline, cost-quality trade-offs) is available in the online supplement at https://llm-uj-research-eval.netlify.app/results_ratings.html.

Results: Critiques & Key Issues

Detailed paper-by-paper critique comparisons (matched issue pairs, severity labels, structural differences) are available in the online supplement at https://llm-uj-research-eval.netlify.app/results_critiques.html.

Supplementary Material

Full LLM reasoning traces, assessment summaries, related work, and extended analyses are available in the online supplement at https://llm-uj-research-eval.netlify.app/appendix_llm_traces.html.

Did Authors Respond to Unjournal Evaluations?

The interactive tables in this appendix are available in the online version at https://valentinklotzbuecher.github.io/llm-uj-research-eval/paper_response_analysis.html.

Note

This page reports an April 2026 tracking snapshot covering 57 Unjournal evaluations. Adjustment status had been assessed for 22 of them, not the full set; completing and re-validating the remaining staff assessment is an ongoing goal. The page also includes an exploratory LLM comparison of eight paper pairs and the more careful July 2026 checks described below. See [Combined Evidence](#).

Comment or flag a correction

This page has Hypothes.is enabled. Select text and open the annotation sidebar from the upper-right edge of the page to leave an inline comment. We especially welcome corrections to document/version matching, missing public revisions, or our interpretation of whether a change reflects a suggestion. A free Hypothes.is account is required to post.

Overview

The Unjournal commissions open, structured evaluations of working papers in economics and social science. A natural question is whether this process changes research: do authors engage with feedback and revise their work?

This page draws on a tracking file covering 57 evaluations. In the April snapshot, 22 had an assessed adjustment status and 35 remained unassessed. For a subset, we also use automated PDF comparison and exploratory LLM screening. Background: [The Unjournal's knowledge base](#).

```
library(dplyr)
library(ggplot2)
library(readr)
library(DT)
library(tidyr)
library(stringr)
```

```

library(glue)
library(jsonlite)

df <- read_csv("data/author_adjustment_manual.csv", show_col_types = FALSE)
reviewed_outcomes <- read_csv(
  "data/paper_response_reviewed_outcomes.csv",
  show_col_types = FALSE
)

stopifnot(
  nrow(reviewed_outcomes) == 22,
  sum(reviewed_outcomes$public_version_status == "no_observed_public_change") == 13,
  sum(reviewed_outcomes$public_version_status == "later_public_version_reviewed") == 9
)

df <- df |>
  rename(
    paper_title      = label_paper_title,
    pubpub_link      = dup_pubpub_final_links,
    adj_status       = `Adjusted paper?`,
    updated_manual    = `Updated since UJ report -- manual confirmation`,
    deposit_after     = `deposit date > unjournal pub date`,
    research_area     = main_cause_cat_abbrev,
    nb_link           = notebookLM_link,
    pub_status        = publication_status,
    wp_date           = working_paper_release_date,
    uj_pub_date       = publication_date_unjournal
  )

adj_labels <- c(
  "Evidence of updating", "Stated intention to update",
  "Mixed / minor updating", "Unlikely to update", "Not yet assessed"
)

df <- df |>
  mutate(
    adj_status_clean = case_when(
      adj_status == "evidence of updating"           ~ "Evidence of updating",
      adj_status == "Stated intention to update"     ~ "Stated intention to update",
      adj_status == "mixed evidence/minor updating" ~ "Mixed / minor updating",
      adj_status == "Unlikely to update"             ~ "Unlikely to update",
      TRUE                                           ~ "Not yet assessed"
    ),
    adj_status_clean = factor(adj_status_clean, levels = adj_labels),
    author_response_clean = case_when(
      author_response == "Formal response" ~ "Formal response",
      author_response == "Informal"       ~ "Informal response",

```

```

    author_response == "None?"          ~ "No response",
    TRUE                               ~ "Not yet coded"
  )
)

theme_uj <- function(base_size = 12) {
  theme_minimal(base_size = base_size) +
    theme(panel.grid.minor = element_blank(),
          axis.text = element_text(size = base_size * 0.85),
          plot.caption = element_text(size = base_size * 0.75, color = "grey50"),
          legend.position = "right")
}

```

Combined Evidence of Paper Updating

Two evidence streams are shown together: April tracking categories for 57 papers, of which 22 had an assessed adjustment status, and an exploratory Claude Opus 4.6 screen for eight papers where before/after PDFs were available and meaningful text changes were detected. The latter is a way to prioritize inspection, not a causal estimate.

! Current validation work (July 2026)

The April figures below remain the historical, partly assessed snapshot. A fresh July 18 source refresh captured 32 public current-paper versions and created 13 distinct before/after comparison records. Nine records passed document, identity, evaluation-mapping, and timing checks. Four superficially usable comparisons were excluded because the supposed later document predates the evaluation. We also audited all 17 cases where the initially selected input and current public PDF were byte-identical. Thirteen support a limited “no observed public document change” finding. Two apparent no-change cases were repaired by locating the correct later endpoint, one revised endpoint remains unavailable, and one record had no completed evaluation. These are coverage figures for a workflow, not findings that an evaluation caused a change.

For each candidate, the revised workflow preserves PDF snapshots and hashes, checks that the documents are the same paper and in the right order, records page- and section-level changes, and requires human review before any public claim of evaluation-linked updating. A newer PDF, a textual difference, or a model suggestion is not enough on its own.

Quick updated takeaways from the stricter screen

- **The substantive rates do not change.** The refresh found two additional different PDFs, but both predate their Unjournal evaluations. The reviewed evidence therefore remains 13 cases with no observed public document change and nine genuine later versions.
- **Whole-document controls matter.** A first card-level pass produced three medium-priority lexical matches. Checking the entire earlier paper downgraded all of them: the final deterministic pass found 17 low-priority review leads and no medium- or high-priority link. This does not overturn the manually reviewed correspondences below; it shows that simple lexical matching should not be used to create them.
- **The most informative low-priority leads illustrate different outcomes.** The macro-climate revision adds an appendix explicitly titled “Additional Robustness Checks,” consistent

with a broad request for more checks but without identifying the evaluation as its source. The resilient-education revision adds questionnaire and protocol material about speakerphone use and other children, which documents the possible spillover issue but does not establish that only one student benefited. An apparent Urban Forests match disappears because alternative upwind-cone checks were already in the earlier appendix.

- **Coverage remains the binding constraint.** Among the 57 tracked records, 21 still need timeline validation and 15 current endpoints failed even after headless-browser fallback. Those cases cannot be treated as either updated or unchanged.

The automated screen is deliberately conservative. It requires an eligible version timeline, exact evaluation-document mapping, an evaluator quote with a valid line anchor, language that is new relative to the aligned change card, a whole-earlier-document preexistence check, and a cross-paper placebo benchmark. Passing produces a review lead only; it never assigns influence or enters the response-rate numerator automatically.

Expanded case review (updated July 18, 2026)

For each of the initial seven cases, separate passes extracted exact suggestions, classified paper changes without seeing the evaluations, tried to disprove suggested links, and adjudicated the surviving candidates. We then manually checked the strongest apparent matches. Across these cases, the process examined 90 mechanically valid suggestion items and 99 changes classified as substantive. For *Urban Forests*, the bounded classifier reviewed 31 of 68 deterministic change cards; 16 of those 31 were substantive. Two additional cases were reviewed after correcting stale endpoint selections.

Paper	Evidence items	Change cards	Current assessment
<i>The Environmental Effects of Economic Production</i>	4	28	One plausible aligned revision, attribution unconfirmed. A new taxa-heterogeneity figure includes birds and partly matches a suggestion to assess productivity effects for bird taxa. The new analysis concerns agricultural income and eight taxa, and there is no author statement connecting it to the evaluation.

Paper	Evidence items	Change cards	Current assessment
<i>Building Resilient Education Systems</i>	10	10	Possible partial correspondence, attribution unconfirmed. A request for stronger robustness checks overlaps with newly reported multiple-testing-adjusted results. Newly appended questionnaire and protocol material also documents speakerphone use and educational activity with other children, but it does not show that only one student benefited per call or rule out spillovers. These are standard design/reporting steps, and no response letter links them to the evaluation.

Paper	Evidence items	Change cards	Current assessment
<i>The Macroeconomic Impact of Climate Change</i>	6	7	Circumstantial correspondence only. The revision adds an “Additional Robustness Checks” appendix and a new cross-sample robustness statement, both topically consistent with the evaluator’s small-sample concern. The request was broad, related robustness material already existed, and no revision note identifies the source.
<i>A Welfare Analysis of Policies Impacting Climate Change</i>	25	6	No supported link at present. One automated candidate appeared to connect a request to clarify “international policies” with international examples. Manual checking found that those examples were already present in the earlier version, so we rejected the match.

Paper	Evidence items	Change cards	Current assessment
<i>Zero-Sum Thinking, the Evolution of Effort-Suppressing Beliefs, and Economic Development</i>	8	21	No positive link found. Three possible pairings reflected timing only; five suggestions remained unmatched.
<i>Water Treatment and Child Mortality</i>	14	11	One plausible aligned revision, attribution unconfirmed. A new formal cost-effectiveness model, including cost per DALY and net DALYs averted, closely matches requests to deepen the cost analysis. No author response or revision memo links the addition to the evaluation.

Paper	Evidence items	Change cards	Current assessment
<i>Urban Forests: Environmental Health Value and Disparities</i>	23	16	Possible topical correspondences, attribution unconfirmed. New wind-direction checks overlap with a confounding concern, and an NDVI footnote partly overlaps with a request about alternative vegetation measures. The strongest automated site-selection match was excluded because its source quote failed validation. A newer lexical match about alternative upwind measures was also downgraded after the whole-document check found alternative upwind-cone tests in the earlier appendix.

Paper	Evidence items	Change cards	Current assessment
<i>Artificial Intelligence and Economic Growth</i>	8	1	Direct documented evaluation-linked correction. The public author response says the evaluation caught a significant proof error. The maintained document marks the exact invalid derivation and points to the corrected proof. The whole-document format change produced only one coarse change card, so this conclusion rests on the explicit response and document annotation rather than model matching.

Paper	Evidence items	Change cards	Current assessment
<i>Adaptability and the Pivot Penalty in Science</i>	8	1	No supported substantive implementation; one weak framing overlap. A manual, section-aligned review found real revisions, but seven of eight suggestions were not implemented. A new caveat that the measured outcomes do not exhaust research quality weakly acknowledges the concern about omitted benefits without adding a benefit outcome. Several apparently responsive items were already in the evaluated version.

The nine reviewed comparisons now include one direct documented correction, two reasonably specific content alignments with attribution unconfirmed, three weaker correspondences, and three cases with no supported substantive match. Only the proof correction has explicit public evidence connecting the evaluation to the change.

Overall and conditional response rates

The 22 reviewed endpoints support a more useful distinction than “changed” versus “unchanged.” The five outcome categories below are mutually exclusive.

Reviewed endpoint outcome	Cases	Share of 22
No observed public document change	13	59%
Later version, but no supported substantive implementation	3	14%

Reviewed endpoint outcome	Cases	Share of 22
Later version with weaker correspondence	3	14%
Later version with a reasonably specific alignment; attribution unconfirmed	2	9%
Explicit public author link between evaluation and correction	1	5%

Percentages are rounded; the counts are primary.

Overall, six of 22 reviewed endpoints (27%) show at least some correspondence between an evaluation point and a later change. Three of 22 (14%) show either a specific alignment or an explicit link, and one of 22 (5%) has an explicit public author statement connecting the evaluation to the correction.

Conditional on the nine papers with a genuine later public version, six (67%) show at least some correspondence, three (33%) show a specific or explicit alignment, and one (11%) has an explicit author link. Three of nine (33%) have no supported substantive implementation in the versions we reviewed.

These are descriptive rates for a small, selected, source-available set, not estimates for all 57 tracked evaluations. “Later version, but no supported implementation” is also not equivalent to “the authors ignored the suggestions.” Authors may have considered and rejected a suggestion, responded privately, revised a document we have not found, or changed the paper for unrelated reasons.

The identity checks now follow normalized DOIs where available and permit explicit, manually reviewed title aliases. They do not depend on exact title equality or infer aliases from a similar title. The evidence record stores the canonical title, any reviewed aliases, and which title matched the PDF; this matters for small changes such as “vs.” to “versus” and for future substantive retitling.

Exploratory differences across research areas

The table counts only the 22 reviewed endpoints. “Any correspondence” includes weak correspondences and therefore should not be read as demonstrated influence.

Tracked research area	Reviewed endpoints	Later public version	Any correspondence overall	Any correspondence, conditional on update
Environment	3	3	2/3	2/3

Tracked research area	Reviewed endpoints	Later public version	Any correspondence overall	Any correspondence, conditional on update
Dev. economics / governance (LMICs)	6	3	2/6	2/3
Social impact of technology / AI	2	1	1/2	1/1
Innovation and meta-science	4	1	0/4	0/1
Global health (LMICs)	6	1	1/6	1/1
Economics, welfare, miscellaneous	1	0	0/1	—

Environment papers were more likely to have a later public version in this reviewed set, while global-health papers were less likely. Conditional response patterns cannot be ranked credibly: most area-specific denominators are one or three updated papers, and case selection partly reflects whether usable public versions could be recovered. The table is best treated as a prompt for stratified sampling as coverage expands.

What kinds of evaluation points aligned with changes?

For an initial qualitative comparison, we assigned one primary evaluation aspect to each of the nine updated cases. This simplifies evaluations that often raised several distinct issues.

Primary evaluation aspect	Cases	Strongest observed result	Pattern in this review
Correctable proof error	1	Explicit documented link	The clearest result was precise, checkable, and acknowledged publicly.
Specified additional analysis	2	Specific alignment, attribution unconfirmed	Both later papers added a closely related analysis, but neither linked it publicly to the evaluation.

Primary evaluation aspect	Cases	Strongest observed result	Pattern in this review
Robustness, confounding, or measurement check	3	Weaker correspondence	Later checks overlapped with the concern, but standard-practice explanations remained plausible.
Broad theory, mechanisms, welfare, framing, or interpretation	3	No supported substantive implementation	The later versions changed, but the broad requested extensions were not clearly carried out.

The tentative pattern is that concrete, falsifiable requests are easier both to act on and to verify. This does not show that they cause more responsiveness: specificity is confounded with feasibility, observability, and the cost of the requested work. A credible next analysis should code all validated suggestions into a common taxonomy, double-code a sample, and compare response rates within updated papers.

Pilot check for later related work

We also ran a targeted search for later work extending three updated papers where the reviewed version showed no supported substantive implementation. This is a separate outcome from revision of the evaluated paper.

Evaluated case	Public continuation checked	Pilot result
<i>A Welfare Analysis of Policies Impacting Climate Change</i>	The July 2026 AER version of record and the October 2025 NBER full text	The AER item is the same paper, not a separate extension. Its full text was not publicly accessible in our check, so the content assessment still rests on the public NBER revision. We did not identify a separate later extension by the author team.
<i>The Pivot Penalty in Research</i>	The May 2025 Nature paper , supplement, and related records	This is the final version of the evaluated research line, not later separate work. It was accepted before the public Unjournal evaluation, and we did not identify a subsequent author-team extension addressing the unmatched suggestions.

Evaluated case	Public continuation checked	Pilot result
<i>Zero-Sum Environments, the Evolution of Effort-Suppressing Beliefs, and Economic Development</i>	The August 2025 NBER revision and targeted title/author searches	The NBER revision is the same paper and is already included in the comparison above. We did not identify clearly attributable later work by the overlapping author team that implements the unmatched suggestions.

This pilot is deliberately phrased as “not identified,” not “does not exist.” Related-work tracking is harder than version tracking: titles, author teams, and empirical settings can all change. A scalable extension should record search dates and queries, require meaningful author overlap and a documented intellectual link, and keep “same paper,” “later extension,” and “unrelated related literature” as separate categories.

Detailed aligned review: *The Pivot Penalty in Research*

The initial comparison was too coarse because the later paper reorganized the manuscript and supplement. We therefore split the evaluated August 2024 bundle at the actual main-text/supplement boundary, aligned later sections by content rather than section number, and checked each validated suggestion against the February 2025 manuscript and the Nature main text and supplement. We also checked the Nature reporting summary, peer-review file, and source-data package for provenance and context; these are not treated as an Unjournal author response.

Evaluation suggestion	What the later documents show	Verdict
Examine promotion or funding outcomes	The evaluated version already checked whether papers acknowledged funding. Later versions do not study promotion or funding received as an outcome.	Not addressed
Account for potential benefits omitted by the outcome measures	Later discussion newly notes that citation, publication, novelty, conventionality, and applied-value measures do not exhaust a multidimensional concept of research quality. It adds no benefit outcome.	Weak framing acknowledgment only
Examine decision-making behind pivoting	The evaluated version already discussed researchers weighing pivots and making sequential choices. Later versions add no choice model or new evidence on the decision process.	Not newly addressed

Evaluation suggestion	What the later documents show	Verdict
Analyze welfare or possible misallocation	Portfolio and pre-positioning policy discussion was already present. No welfare or misallocation analysis was added.	Not addressed
Test longer horizons, including the COVID/mRNA setting	Alternative citation windows and time-trend figures were already present. No new COVID/mRNA long-horizon test was added.	Not addressed as requested
Test when in a career or sequence researchers should pivot	Existing career-stage and sequential-dynamics material does not provide the requested timing-heterogeneity test.	Not addressed
Adjust for evolving journal preferences	We found no aggregate journal-preference adjustment in the later documents.	Not addressed
Define paper pivots by fields rather than journals	The evaluated supplement already reconstructed the paper-pivot measure using L1 fields and still found a penalty. Later versions retain that robustness check and keep journals as the main definition.	Already present; main choice unchanged

The later documents do contain substantive revisions: an updated retraction sample, explicit COVID sample sizes and effect changes, reorganized methods and supplementary material, improved replication links, and the new multidimensional-quality caveat. The principal estimates and conclusions remain stable. The February manuscript predates both Nature acceptance and the public Unjournal evaluation; the Nature paper was accepted on April 18, 2025, while the public evaluation is dated May 21. There is no posted Unjournal author response. Without private transmission dates, none of these revisions should be attributed to the evaluation. We also lack the separate February supplementary files, so supplementary-material changes can be established only from August 2024 to the Nature version.

Audit of byte-identical cases (updated July 18, 2026)

An identical current PDF is potentially informative, but only if the local input is credibly the document that was evaluated. Repository history was not enough: all 17 local inputs entered this repository after their evaluations. We therefore checked official version histories, archived pre-evaluation PDFs, distinctive page-level references in the evaluations, and fresh publisher downloads.

Thirteen cases now support the limited statement **“no observed public document change as of July 18, 2026.”** This does not imply that authors ignored the evaluation. They may have revised another manuscript, made changes that are not publicly available, or decided that no revision was

warranted.

Paper	Evidence binding the evaluated and current documents
<i>Pharmaceutical Pricing and R&D as a Global Public Good</i>	The official NBER record retains its May 2023 issue date with no revision date; a fresh publisher download has the same hash.
<i>Towards Best Practices in AGI Safety and Governance</i>	arXiv lists only version 1 (May 11, 2023); the current PDF is byte-identical.
<i>Choose Your Moments: NIH Peer Review and Scientific Risk Taking</i>	The official NBER record retains its June 2023 issue date with no revision date; a fresh publisher download has the same hash.
<i>The Returns to Science in the Presence of Technological Risks</i>	The evaluation identifies arXiv version 2 (May 9, 2024); arXiv lists no later version.
<i>Effects of Emigration on Rural Labor Markets</i>	The evaluation quotes distinctive page-level text in the October 2017 NBER PDF; the current file has the same hash.
<i>Legalizing Entrepreneurship</i>	The evaluation cites distinctive page-specific text, including an unusual typo, in the March 2023 revision; the current file has the same hash.
<i>The Benefits and Costs of Guest Worker Programs</i>	A pre-evaluation Wayback capture and the current NBER PDF are byte-identical.
<i>The Comparative Impact of Cash Transfers and a Psychotherapy Program</i>	A pre-evaluation Wayback capture and the current NBER PDF are byte-identical.
<i>Maternal Cash Transfers for Child Development</i>	A distinctive page-level quotation binds the evaluation to the August 2025 NBER revision; the current publisher PDF has the same hash.
<i>Stagnation and Scientific Incentives</i>	A pre-evaluation Wayback capture and the current NBER PDF are byte-identical.
<i>Intergenerational Child Mortality Impacts of Deworming</i>	Distinctive page and table references bind the evaluation to the original April 2023 NBER PDF; NBER lists no later revision and the current file has the same hash.
<i>Economic versus Epidemiological Approaches to Policy Evaluation</i>	A distinctive figure-level comparison binds the evaluation to the July 2022 NBER PDF; NBER lists no later revision and the current file has the same hash.
<i>The Long-Run Effects of Psychotherapy on Depression, Beliefs, and Economic Outcomes</i>	A pre-evaluation Wayback capture and a version-specific appendix reference bind the evaluation to the May 2022 NBER PDF; the current NBER file is byte-identical. An author-hosted January 2023 draft was ruled out because its corresponding appendix table contains a different analysis.

The other four initially identical pairs should not be counted as no-change findings. Two have now

been repaired by selecting the author-shared or corrected endpoint and appear in the comparison table above. One paper is known to have been revised, but the revised manuscript is not publicly available; one record had no completed evaluation. These failures show why a hash comparison without source-history checks can give the wrong answer.

i Pipeline and caveats

Current scripts: `scripts/paper_response_evidence.py` for source snapshots, validation, and comparison cards; `scripts/screen_paper_response_matches.py` for deterministic, placebo-benchmarked suggestion-to-change leads; `scripts/run_paper_response_agents.py` for optional bounded, review-only model triage.

- **Historical screen shown below:** before version = PDF sent to evaluators; after version = then-current NBER/arXiv PDF, fetched April 2026. Eight pairs had meaningful text changes and a matched evaluation file.
- **Current validation:** the refresh keeps immutable PDF snapshots; verifies paper identity and chronology; produces page- and section-anchored change cards; and puts ambiguous cases into human review. A transient endpoint failure now preserves the last captured comparison with an explicit cached-snapshot warning instead of erasing it from the queue.
- **Automated correspondence check:** suggestion cues must have valid source anchors. Matches are discounted when the same concepts occur in a local window anywhere in the earlier paper and are benchmarked against matches to other papers' changes. The output is a priority queue, not a response label.
- **Models:** the historical Opus screen used bounded extracted text, so it could not reliably assess figures, tables, or all document context. The current agent triage is also deliberately non-decisive: it can identify evidence to inspect, but cannot label causal influence.
- **Interpretation:** a newer version does not imply that changes were evaluation-driven. Some papers are already being revised, and a substantial revision can have unrelated causes.

```
nn <- function(x, default = NA) if (!is.null(x)) x else default

llm_results_path <- "data/paper_change_llm_results.json"

# Signal tier labels used throughout - define once
tier_levels <- c(
  "Confirmed, LLM >=40%",
  "Confirmed, LLM <40%",
  "LLM >=30% (not confirmed)",
  "LLM 10-29%",
  "LLM <10%"
)

tier_colors_llm <- c(
  "Confirmed, LLM >=40%"      = "#1a7a47",
  "Confirmed, LLM <40%"      = "#2D9D5E",
  "LLM >=30% (not confirmed)" = "#5AE08A",
```

```

"LLM 10-29%"          = "#a8d5b5",
"LLM <10%"            = "#d0e8d8"
)

if (file.exists(llm_results_path)) {
  llm_raw <- fromJSON(llm_results_path, simplifyVector = FALSE)

  llm_df <- lapply(llm_raw, function(r) {
    oa <- r$overall_assessment
    data.frame(
      paper_title      = nn(r$paper_title, ""),
      adj_status_csv   = nn(r$adj_status, ""),
      deposit_after_uj = isTRUE(r$deposit_after_uj),
      text_chg_pct     = nn(r$text_length_change_pct, NA_real_),
      additions        = nn(r$additions_count, NA_integer_),
      deletions        = nn(r$deletions_count, NA_integer_),
      n_major_changes  = length(nn(r$major_changes, list())),
      n_suggestions     = length(nn(r$evaluator_suggestions, list())),
      pct_influenced    = if (!is.null(oa)) nn(oa$pct_likely_influenced, NA_real_) else NA_real_,
      attr_confidence   = if (!is.null(oa)) nn(oa$confidence, NA_real_) else NA_real_,
      narrative         = if (!is.null(oa)) nn(oa$narrative, "") else "",
      eval_found        = !is.null(r$eval_file) && nchar(nn(r$eval_file, "")) > 0,
      skipped           = !is.null(r$skipped_reason),
      stringsAsFactors = FALSE
    )
  }) |> bind_rows()

  llm_analyzed <- llm_df |>
    filter(!skipped & eval_found & n_major_changes > 0 & !is.na(pct_influenced))

  llm_enriched <- llm_analyzed |>
    left_join(df |> select(paper_title, research_area, pubpub_link), by = "paper_title") |>
    mutate(
      manual_tier = case_when(
        adj_status_csv == "evidence of updating" ~ "Confirmed update",
        adj_status_csv == "Stated intention to update" ~ "Stated intention",
        adj_status_csv == "mixed evidence/minor updating" ~ "Mixed / minor",
        adj_status_csv == "Unlikely to update" ~ "Unlikely",
        TRUE ~ "Unclassified"
      ),
      combined_tier = case_when(
        manual_tier == "Confirmed update" & pct_influenced >= 40 ~ tier_levels[1],
        manual_tier == "Confirmed update" ~ tier_levels[2],
        pct_influenced >= 30 ~ tier_levels[3],
        pct_influenced >= 10 ~ tier_levels[4],
        TRUE ~ tier_levels[5]
      ),

```

```

    combined_tier = factor(combined_tier, levels = tier_levels)
  )

  n_analyzed      <- nrow(llm_enriched)
  n_total_fetched <- nrow(llm_df)
  med_pct         <- median(llm_enriched$pct_influenced, na.rm = TRUE)
  n_high          <- sum(llm_enriched$pct_influenced >= 30, na.rm = TRUE)
} else {
  llm_df <- llm_analyzed <- llm_enriched <- data.frame()
  n_analyzed <- n_total_fetched <- n_high <- 0
  med_pct <- NA_real_
}

if (nrow(llm_enriched) > 0) {
  n_confirmed <- sum(llm_enriched$manual_tier == "Confirmed update")
  n_llm_high  <- sum(llm_enriched$manual_tier == "Unclassified" &
                    llm_enriched$pct_influenced >= 30)
  cat(glue::glue(
    "Of {n_total_fetched} papers with fetchable DOIs, {n_analyzed} had meaningful text ",
    "changes and a matched evaluation. {n_confirmed} are manually confirmed updates; ",
    "of the remaining {n_analyzed - n_confirmed}, the LLM finds {n_llm_high} with at least 30%
    "of changes likely driven by evaluator feedback (median across all: {round(med_pct)}%).\"
  ))
}

```

Of 22 papers with fetchable DOIs, 8 had meaningful text changes and a matched evaluation. 3 are manually confirmed updates; of the remaining 5, the LLM finds 2 with at least 30% of changes likely driven by evaluator feedback (median across all: 22%).

```

if (nrow(llm_enriched) > 0) {
  # Build one row per paper with tier assignment
  waffle_llm <- llm_enriched |>
    select(paper_title, combined_tier) |>
    mutate(group = as.character(combined_tier))

  waffle_manual <- df |>
    filter(!paper_title %in% llm_enriched$paper_title) |>
    mutate(group = case_when(
      adj_status_clean == "Evidence of updating"      ~ "Confirmed (manual only)",
      adj_status_clean == "Stated intention to update" ~ "Stated intention",
      adj_status_clean == "Mixed / minor updating"    ~ "Mixed / minor",
      adj_status_clean == "Unlikely to update"        ~ "Unlikely",
      TRUE                                             ~ "Not yet assessed"
    )) |>
    select(paper_title, group)

  waffle_all_groups <- c(tier_levels,
    "Confirmed (manual only)", "Stated intention",

```

```

    "Mixed / minor", "Unlikely", "Not yet assessed")

waffle_colors <- c(
  tier_colors_llm,
  "Confirmed (manual only)" = "#2B7CE9",
  "Stated intention"        = "#7EB8F7",
  "Mixed / minor"           = "#E8722A",
  "Unlikely"                 = "#C45A1E",
  "Not yet assessed"        = "#CBD5E1"
)

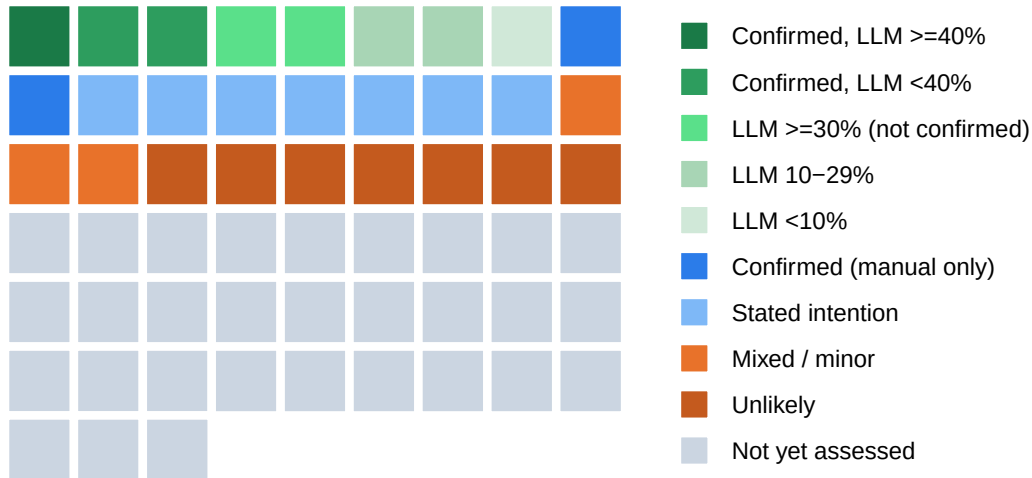
waffle_data <- bind_rows(waffle_llm, waffle_manual) |>
  mutate(group = factor(group, levels = waffle_all_groups)) |>
  arrange(group) |>
  mutate(rank = row_number(),
         x = (rank - 1) %% 9,
         y = -((rank - 1) %/% 9))

ggplot(waffle_data, aes(x = x, y = y, fill = group)) +
  geom_tile(color = "white", linewidth = 1.5) +
  coord_equal() +
  scale_fill_manual(values = waffle_colors, name = NULL,
                    drop = FALSE) +
  theme_void() +
  theme(legend.position = "right",
        legend.text = element_text(size = 9),
        plot.title = element_text(size = 11, face = "plain",
                                   margin = margin(b = 6)),
        plot.caption = element_text(size = 8, color = "grey40",
                                   hjust = 0, margin = margin(t = 8))) +
  labs(title = "All 57 tracked papers - each square = 1 paper",
        caption = "LLM % = share of major post-evaluation changes attributed to evaluator feedback")
}

```

Figure 4: April tracking snapshot for all 57 papers (each square = 1 paper); adjustment status had been assessed for 22. Green tones: papers where LLM analysis was run, shaded by attribution score and recorded confirmation status. Blue: recorded updates without LLM analysis. Orange/grey: stated intention, weak signal, or not yet assessed.

All 57 tracked papers – each square = 1 paper

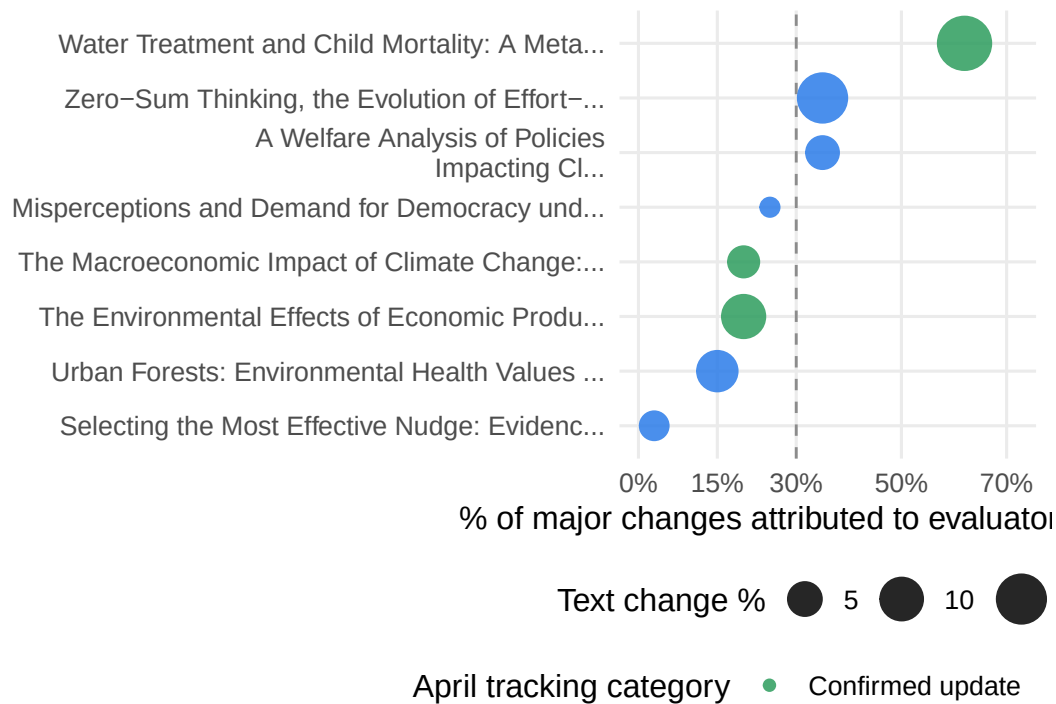


LLM % = share of major post-evaluation changes attributed to evaluator feedback (Claude Opus 4.6)

```
if (nrow(llm_enriched) > 0) {
  dot_colors <- c("Confirmed update" = "#2D9D5E", "Unclassified" = "#2B7CE9")

  llm_enriched |>
    mutate(paper_short = str_trunc(paper_title, 46)) |>
    ggplot(aes(x = pct_influenced, y = reorder(paper_short, pct_influenced),
              color = manual_tier, size = text_chg_pct)) +
    geom_vline(xintercept = 30, linetype = "dashed", color = "grey55", linewidth = 0.5) +
    geom_point(alpha = 0.85) +
    scale_color_manual(values = dot_colors, name = "April tracking category") +
    scale_size_continuous(range = c(3, 9), name = "Text change %") +
    scale_x_continuous(limits = c(0, 72), labels = \(x) paste0(x, "%"),
                       breaks = c(0, 15, 30, 50, 70)) +
    labs(x = "% of major changes attributed to evaluator feedback", y = NULL) +
    theme_uj() +
    theme(legend.position = "bottom", legend.box = "vertical",
          legend.margin = margin(0, 0, 0, 0))
}
```

Figure 5: LLM attribution for the 8 papers with pre/post versions compared. x-axis: estimated % of major changes attributed to evaluator feedback (Claude Opus 4.6, conservative). Point size scales with text change magnitude. Green = manually confirmed update; blue = not yet manually classified. Dashed line at 30%.



Summary Statistics

```
n_total      <- nrow(df)
n_assessed   <- df |> filter(adj_status_clean != "Not yet assessed") |> nrow()
n_evidence   <- df |> filter(adj_status_clean == "Evidence of updating") |> nrow()
n_intention  <- df |> filter(adj_status_clean == "Stated intention to update") |> nrow()
n_any_positive <- df |>
  filter(adj_status_clean %in% c("Evidence of updating",
                                "Stated intention to update",
                                "Mixed / minor updating")) |> nrow()
n_formal     <- df |> filter(author_response_clean == "Formal response") |> nrow()
n_any_resp   <- df |>
  filter(author_response_clean %in% c("Formal response", "Informal response")) |> nrow()
```

Of 57 tracked evaluations, 22 have an assessed adjustment status. Among these: 5 are coded as showing clear evidence of substantive updating, 7 as stating an intention to update, and 15 as showing at least some positive signal. Response status is recorded for only part of the set: 19 papers have a recorded formal or informal response, of which 16 are formal written responses.

```
resp_colors <- c(
  "Formal response" = "#2B7CE9",
  "Informal response" = "#5DA3F5",
  "No response" = "#CBD5E1",
  "Not yet coded" = "#E2E8F0")
```

Table 13: Papers with LLM attribution results, sorted by % of changes attributed to evaluator feedback. ‘Confirmed’ = manually verified by Unjournal staff. Signal tier: ‘Confirmed, LLM $\geq 40\%$ ’ = manually confirmed AND LLM finds strong corroboration; ‘Confirmed, LLM $< 40\%$ ’ = confirmed but weaker LLM signal; ‘LLM $\geq 30\%$ (not confirmed)’ = unconfirmed but LLM finds likely influence.

```

if (nrow(llm_enriched) > 0) {
  tier_bg <- c(
    "Confirmed, LLM \u226540%"      = "#c8e6c9",
    "Confirmed, LLM <40%"          = "#d4edda",
    "LLM \u226530% (not confirmed)" = "#e8f5e9",
    "LLM 10\u201332%"              = "#fff3e0",
    "LLM <10%"                    = "#f8fadc"
  )

  tbl_llm <- llm_enriched |>
    arrange(desc(pct_influenced)) |>
    mutate(
      Title = ifelse(
        !is.na(pubpub_link) & nchar(pubpub_link) > 0,
        paste0('<a href="', pubpub_link, '" target="_blank">',
              str_trunc(paper_title, 60), '</a>'),
        str_trunc(paper_title, 60)
      ),
      `Manual`      = manual_tier,
      `LLM %`       = paste0(round(pct_influenced), "%"),
      `Conf.`       = paste0(attr_confidence, "/5"),
      `Signal tier`  = combined_tier,
      `LLM narrative` = str_trunc(narrative, 350)
    ) |>
    select(Title, Manual, `LLM %`, Conf., `Signal tier`, `LLM narrative`)

  datatable(
    tbl_llm,
    escape      = FALSE,
    rownames    = FALSE,
    options     = list(
      pageLength = 5,
      dom        = "tip",
      scrollX     = TRUE,
      columnDefs = list(
        list(width = "18%", targets = 0),
        list(width = "11%", targets = 1),
        list(width = "6%",  targets = 2),
        list(width = "5%",  targets = 3),
        list(width = "16%", targets = 4),
        list(width = "44%", targets = 5)
      )
    )
  ) |>
  formatStyle("Signal tier",
    backgroundColor = styleEqual(names(tier_bg), unname(tier_bg)))
} else {
  cat("*No LLM attribution results available yet.*")
}

```

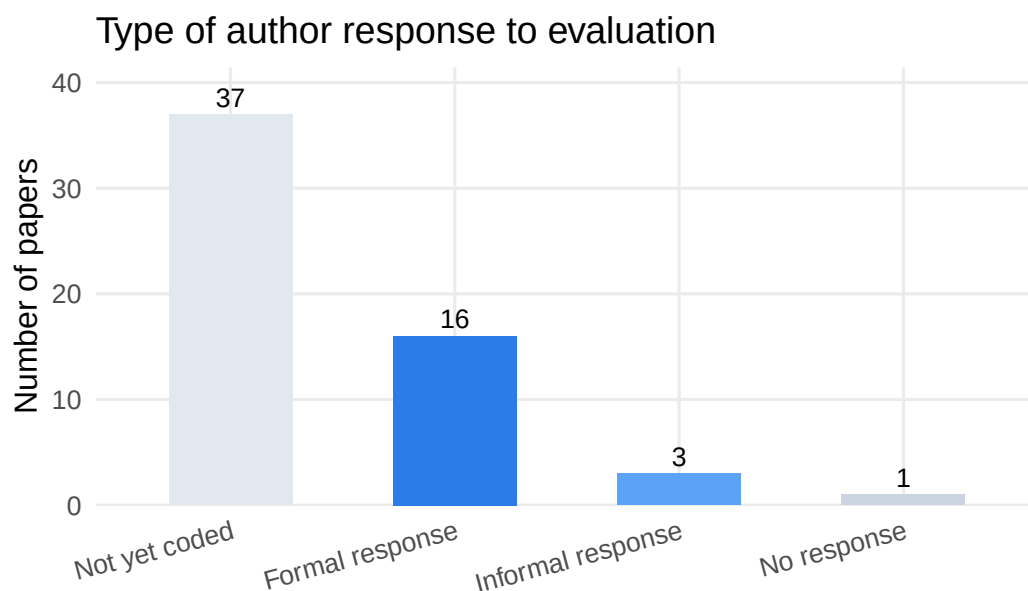
```

)

df |>
  count(author_response_clean) |>
  ggplot(aes(x = reorder(author_response_clean, -n), y = n,
                        fill = author_response_clean)) +
  geom_col(width = 0.55, show.legend = FALSE) +
  geom_text(aes(label = n), vjust = -0.4, size = 3.5) +
  scale_fill_manual(values = resp_colors) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.12))) +
  labs(x = NULL, y = "Number of papers",
       title = "Type of author response to evaluation") +
  theme_uj() +
  theme(axis.text.x = element_text(angle = 15, hjust = 1))

```

Figure 6: Type of author response to Unjournal evaluation. ‘Not yet coded’ indicates the response status has not been assessed for those papers (37 blanks in data); this is distinct from confirmed non-response (only 1 paper is explicitly coded as no response).



Full Tabulation

i All 57 tracked evaluations

Click column headers to sort; use the search box to filter. Adjustment status is color-coded: green = positive signal, orange = minor/uncertain, grey = unlikely or not yet assessed.

Table 14

```
tbl <- df |>
  mutate(
    title_linked = if_else(
      !is.na(pubpub_link) & pubpub_link != "",
      paste0('<a href="", pubpub_link, '" target="_blank">',
            str_trunc(paper_title, 72), '</a>'),
      str_trunc(paper_title, 72)
    ),
    authors_short = str_trunc(authors, 40),
    nb_icon = if_else(
      !is.na(nb_link) & nb_link != "",
      paste0('<a href="", nb_link, '" target="_blank" title="NotebookLM">&#128214;</a>'),
      ""
    )
  ) |>
  select(Paper = title_linked, Authors = authors_short,
         Area = research_area, Response = author_response_clean,
         Adjustment = adj_status_clean, `Pub.` = pub_status, NB = nb_icon)

adj_bg <- c(
  "Evidence of updating"      = "#d4edda",
  "Stated intention to update" = "#e8f5e9",
  "Mixed / minor updating"    = "#fff3e0",
  "Unlikely to update"        = "#fce4d4",
  "Not yet assessed"          = "#f8fafc"
)

datatable(tbl, escape = FALSE, rownames = FALSE, filter = "top",
  options = list(
    pageLength = 8, autoWidth = FALSE, scrollX = TRUE, dom = "lfrtip",
    columnDefs = list(
      list(width = "36%", targets = 0),
      list(width = "18%", targets = 1),
      list(width = "8%", targets = 2),
      list(width = "11%", targets = 3),
      list(width = "16%", targets = 4),
      list(width = "8%", targets = 5),
      list(width = "3%", targets = 6)
    )
  ),
  caption = "All tracked Unjournal evaluations. NB = NotebookLM analysis link."
) |>
  formatStyle("Adjustment",
    backgroundColor = styleEqual(names(adj_bg), unname(adj_bg)))
```